

(54) **RECIPIENT-SPECIFIC VOICE TONE ADJUSTMENT IN TELEPHONY**

(71) Applicant: **International Business Machines Corporation**, Armonk, NY (US)

(72) Inventors: **Guang Han Sui**, BEIJING (CN); **Peng Hui Jiang**, BEIJING (CN); **Su Liu**, Austin, TX (US); **Jun Su**, BEIJING (CN)

(73) Assignee: **International Business Machines Corporation**, Armonk, NY (US)

(21) Appl. No.: **18/599,431**

(22) Filed: **Mar. 8, 2024**

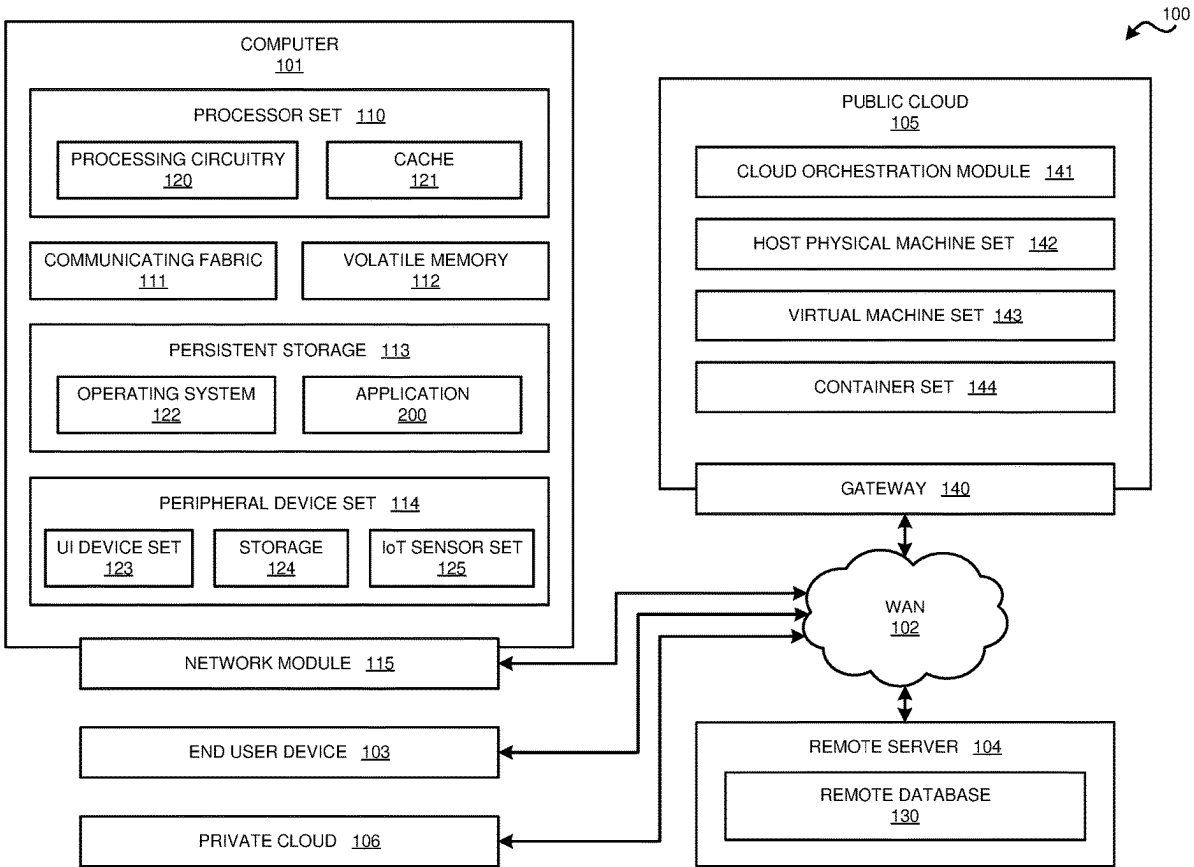
Publication Classification

(51) **Int. Cl.**
G10L 13/08 (2013.01)
G10L 15/02 (2006.01)
G10L 15/26 (2006.01)

(52) **U.S. Cl.**
CPC **G10L 13/08** (2013.01); **G10L 15/02** (2013.01); **G10L 15/26** (2013.01)

(57) **ABSTRACT**

An embodiment extracts, from a plurality of voice samples, voice tone data. The embodiment converts, using a speech to text model, a speech input to corresponding text. The embodiment generates a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data.



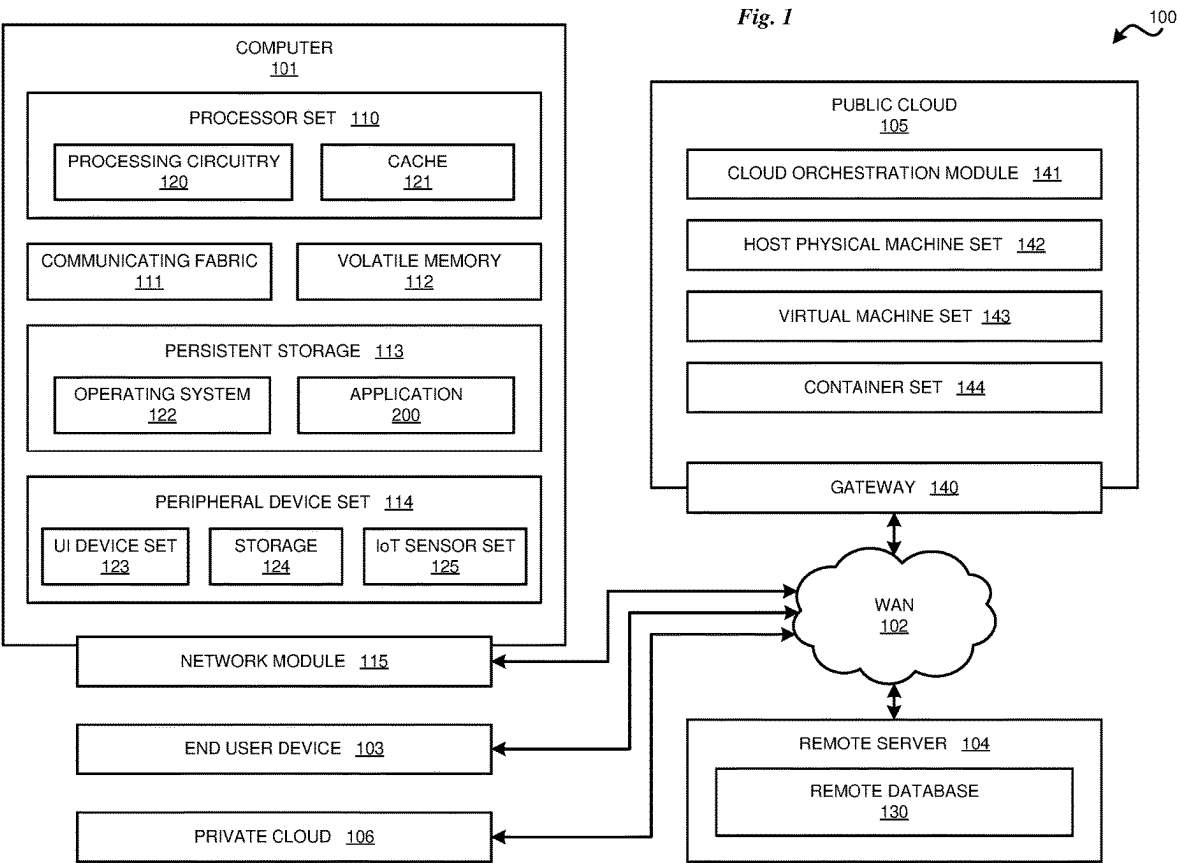


Fig. 2

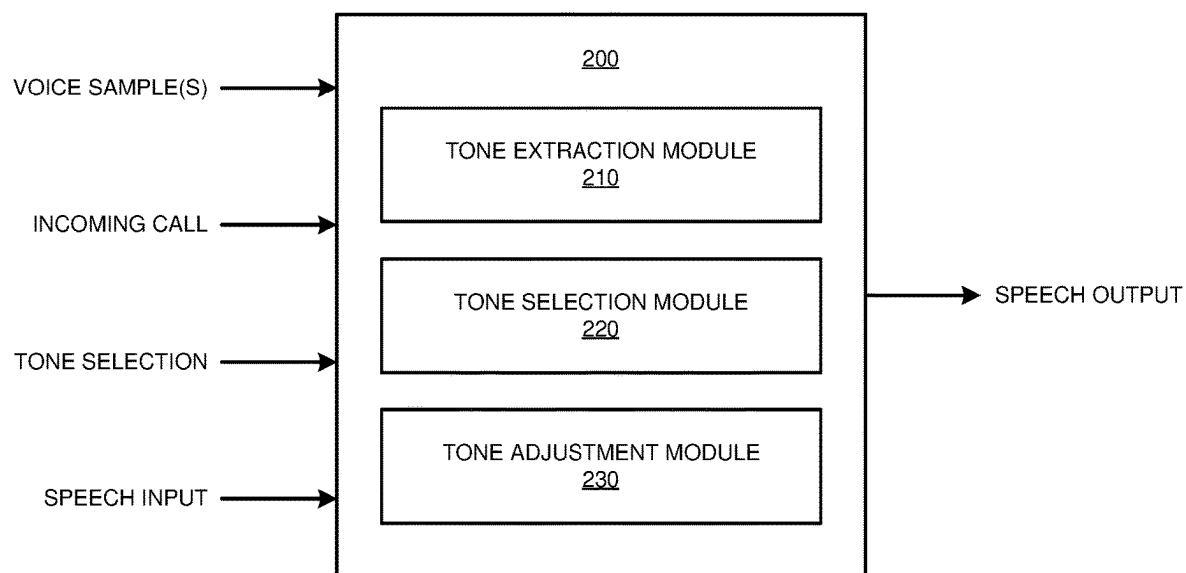


Fig. 3

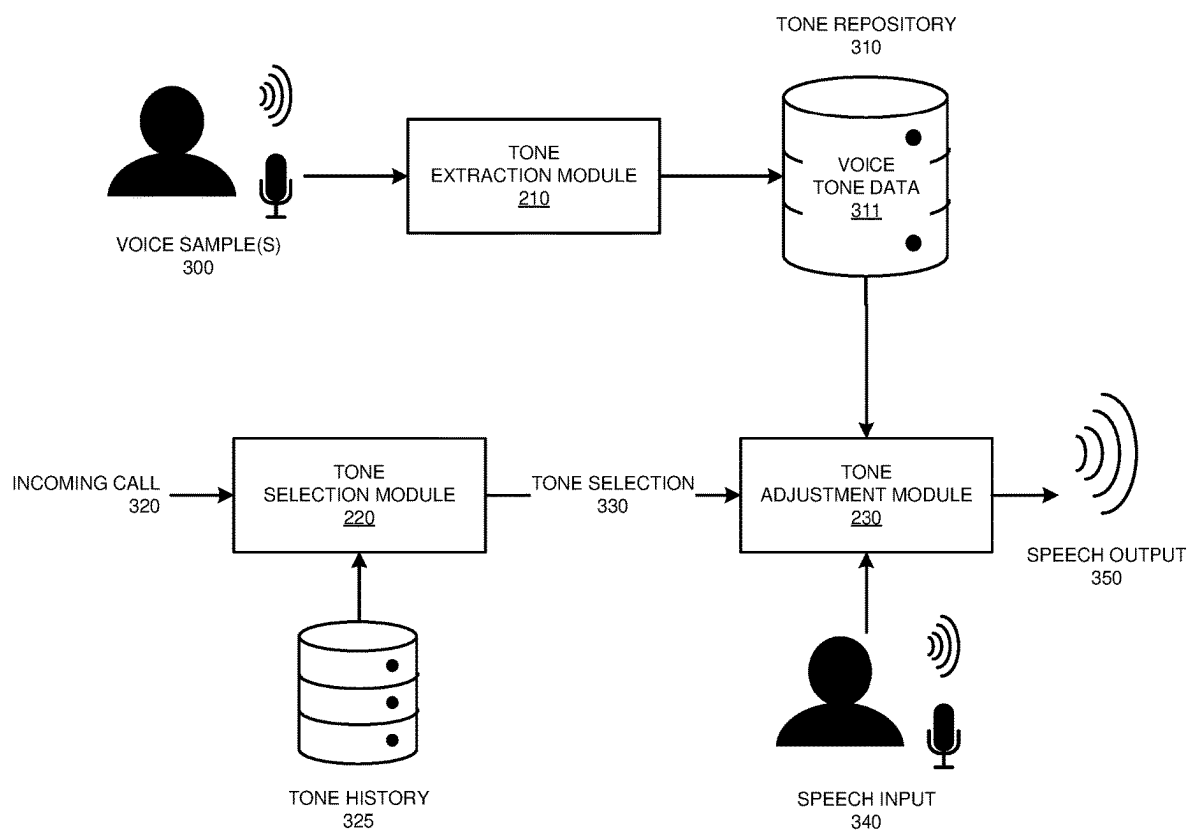
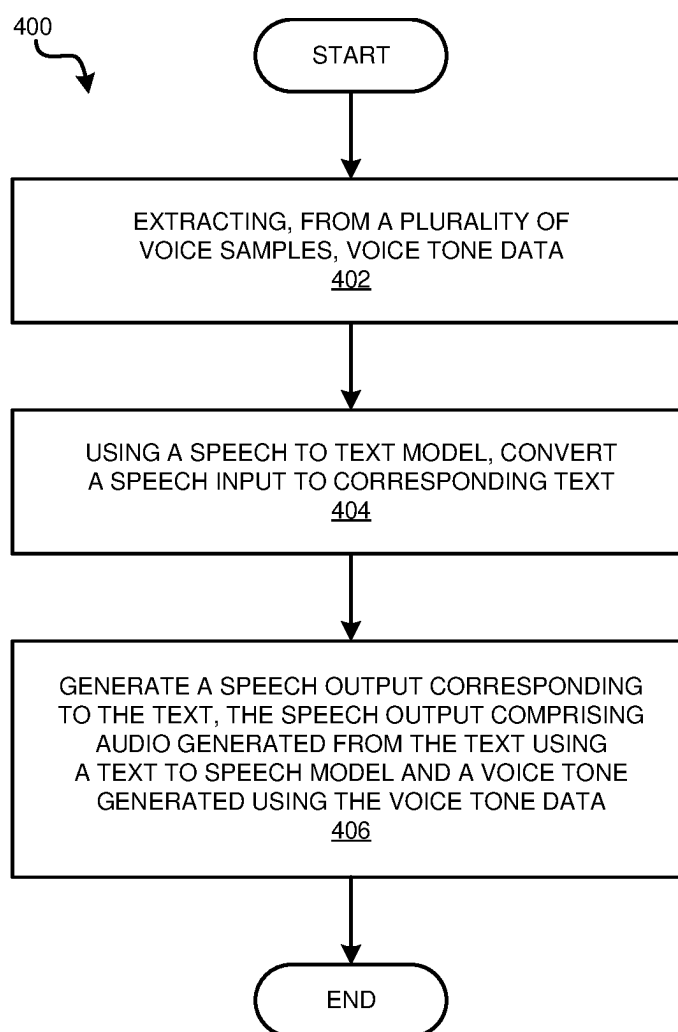


Fig. 4



RECIPIENT-SPECIFIC VOICE TONE ADJUSTMENT IN TELEPHONY

BACKGROUND

[0001] The present invention relates generally to voice communication technology. More particularly, the present invention relates to a method, system, and computer program for recipient-specific voice tone adjustment in telephony.

[0002] Voice tone, or tone of voice, or tone, as used herein, refers to the quality, manner or way an individual speaks. Some aspects of voice tone are a person's pitch and variations in pitch, speed, rhythm, melody, and accent. For example, in customer service one desired tone of voice might be warm and enthusiastic, characterized by a comparatively higher-pitched voice, some variation in pitch while speaking, speaking quickly, and conveying a friendly, upbeat attitude, while another tone of voice might be cold and authoritative, using a deeper, lower-pitched voice and less variation in pitch, and speaking more slowly, to convey seriousness and firmness, and a third tone of voice might be warm and professional, striking a balance between friendliness and seriousness.

SUMMARY

[0003] The illustrative embodiments provide for recipient-specific voice tone adjustment in telephony. An embodiment includes extracting, from a plurality of voice samples, voice tone data. The embodiment includes converting, using a speech to text model, a speech input to corresponding text. The embodiment includes generating a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the embodiment.

[0004] An embodiment includes a computer usable program product. The computer usable program product includes a computer-readable storage medium, and program instructions stored on the storage medium.

[0005] An embodiment includes a computer system. The computer system includes a processor, a computer-readable memory, and a computer-readable storage medium, and program instructions stored on the storage medium for execution by the processor via the memory.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The novel features believed characteristic of the invention are set forth in the appended claims. The invention itself, however, as well as a preferred mode of use, further objectives, and advantages thereof, will best be understood by reference to the following detailed description of the illustrative embodiments when read in conjunction with the accompanying drawings, wherein:

[0007] FIG. 1 depicts a block diagram of a computing environment in accordance with an illustrative embodiment;

[0008] FIG. 2 depicts a block diagram of an example configuration for recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment;

[0009] FIG. 3 depicts an example of recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment; and

[0010] FIG. 4 depicts a flowchart of an example process for recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

[0011] The illustrative embodiments recognize that people often use different tones of voice when speaking with different people or in different contexts. For example, one might use a higher-pitched voice, some variation in pitch while speaking, and speaking quickly when talking to a close friend or family member, use a deeper, lower-pitched voice and less variation in pitch, and speak more slowly, to convey seriousness and firmness when speaking with a subordinate in a work environment, or use a combination to convey warmth and professionalism when interacting with a co-worker or a customer. However, selecting the correct tone of voice when communicating over a device (e.g., a dedicated telephone, a voice communication application executing on a device, or an audio or video conference system or application executing on a device) can be difficult. For incoming calls, caller identification might provide only a telephone number which a user might not recognize, or provide a generic caller name such as "wireless caller". As well, caller identification is not always available, and a user might not look at the caller identification before answering a call. For outgoing calls, a user might have only a telephone number, with little or no information as to who the person answering is or which tone of voice to select. Also, work requirements might dictate selection of a particular voice tone during voice communication. For example, an employer might want all of its customer service representatives to use a particular tone of voice. However, tone of voice training can be expensive and insufficiently effective for some. Thus, the illustrative embodiments recognize that there is an unmet need to implement voice tone adjustment technologically, when communicating over a device, in a recipient-specific manner.

[0012] The present disclosure addresses the deficiencies described above by providing a process (as well as a system, method, machine-readable medium, etc.) that extracts voice tone data from a plurality of voice samples, uses a speech to text model to convert a speech input to corresponding text, and generates a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data. Thus, the illustrative embodiments provide for recipient-specific voice tone adjustment in telephony.

[0013] Though this disclosure pertains to the collection of personal data (e.g., speech data), it is noted that in embodiments, users opt-in to the system. In doing so, they are informed of what data is collected and how it will be used, that any collected personal data may be encrypted while being used, that users can opt-out at any time, and that if they opt-out, any personal data of the user is deleted.

[0014] An illustrative embodiment receives a plurality of voice samples of a user speaking in a specified voice tone. An embodiment uses a presently available technique to extract voice tone data from the plurality of voice samples. The voice tone data includes data usable to generate the specified voice tone. Some non-limiting examples of pres-

ently available techniques to extract voice tone data are a voice pitch analyzer which determines voice pitch (i.e., frequency) and variations in pitch, and machine learning models trained to identify emotions such as happiness, sadness, anger, or excitement within a plurality of voice samples. A user has the option to save or discard voice tone data usable to generate a specified voice tone, to associate a specific recipient for a specific saved voice tone, and to set a default saved voice tone. For example, an embodiment might receive a plurality of voice samples of a user speaking in a warm voice tone, extract voice tone data from the voice samples, allow the user to save the voice tone data as “family”, and associate the “family” voice tone data with contact information for the user’s spouse. As another example, an embodiment might receive a plurality of voice samples of a user speaking in a cool, formal voice tone, extract voice tone data from the voice samples, allow the user to save the voice tone data as “customer”, and designate the “customer” voice tone data as the default voice tone. One embodiment saves, maintains, and manages voice tone data in a voice tone repository specific to a user.

[0015] An embodiment offers a user an option to select a saved voice tone for use during a voice communication (e.g., an incoming or outgoing audio call or audio or video conferencing session). A user saves a user’s voice tone selection for use in an ongoing voice communication, as well as in future voice communications with the same recipient as the current voice communication. For example, if the user selects the saved “family” voice tone for use in a call with a device designated in the user’s contacts as the user’s spouse, an embodiment uses the saved “family” voice tone in the user’s next voice communication with his or her spouse. Another embodiment selects a saved voice tone for use during a particular voice communication automatically, using a previously-saved voice tone or a default voice tone.

[0016] During a voice communication (e.g., an audio call or audio or video conferencing session) in which a saved voice tone has been selected, an embodiment uses a presently available speech to text model to convert the user’s speech input to corresponding text, and generates a speech output corresponding to the text. The speech output includes audio generated from the text using a presently available text to speech model and a voice tone generated using the saved voice tone data and a presently available technique. Two non-limiting examples of presently available techniques are Microsoft VALL-E (a language modeling approach for text-to-speech synthesis that includes voice tone support) and Apple Personal Voice. The intermediate text is necessary as currently available voice tone generation techniques operate on text rather than speech in audio form.

[0017] For the sake of clarity of the description, and without implying any limitation thereto, the illustrative embodiments are described using some example configurations. From this disclosure, those of ordinary skill in the art will be able to conceive many alterations, adaptations, and modifications of a described configuration for achieving a described purpose, and the same are contemplated within the scope of the illustrative embodiments.

[0018] Furthermore, simplified diagrams of the data processing environments are used in the figures and the illustrative embodiments. In an actual computing environment, additional structures or components that are not shown or described herein, or structures or components different from

those shown but for a similar function as described herein may be present without departing the scope of the illustrative embodiments.

[0019] Furthermore, the illustrative embodiments are described with respect to specific actual or hypothetical components only as examples. Any specific manifestations of these and other similar artifacts are not intended to be limiting to the invention. Any suitable manifestation of these and other similar artifacts can be selected within the scope of the illustrative embodiments.

[0020] The examples in this disclosure are used only for the clarity of the description and are not limiting to the illustrative embodiments. Any advantages listed herein are only examples and are not intended to be limiting to the illustrative embodiments. Additional or different advantages may be realized by specific illustrative embodiments. Furthermore, a particular illustrative embodiment may have some, all, or none of the advantages listed above.

[0021] Furthermore, the illustrative embodiments may be implemented with respect to any type of data, data source, or access to a data source over a data network. Any type of data storage device may provide the data to an embodiment of the invention, either locally at a data processing system or over a data network, within the scope of the invention. Where an embodiment is described using a mobile device, any type of data storage device suitable for use with the mobile device may provide the data to such embodiment, either locally at the mobile device or over a data network, within the scope of the illustrative embodiments.

[0022] The illustrative embodiments are described using specific code, computer readable storage media, high-level features, designs, architectures, protocols, layouts, schematics, and tools only as examples and are not limiting to the illustrative embodiments. Furthermore, the illustrative embodiments are described in some instances using particular software, tools, and data processing environments only as an example for the clarity of the description. The illustrative embodiments may be used in conjunction with other comparable or similarly purposed structures, systems, applications, or architectures. For example, other comparable mobile devices, structures, systems, applications, or architectures therefor, may be used in conjunction with such embodiment of the invention within the scope of the invention. An illustrative embodiment may be implemented in hardware, software, or a combination thereof.

[0023] The examples in this disclosure are used only for the clarity of the description and are not limiting to the illustrative embodiments. Additional data, operations, actions, tasks, activities, and manipulations will be conceivable from this disclosure and the same are contemplated within the scope of the illustrative embodiments.

[0024] Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

[0025] A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

[0026] With reference to FIG. 1, this figure depicts a block diagram of a computing environment 100. Computing environment 100 contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as application 200 implementing recipient-specific voice tone adjustment in telephony. In addition to block 200, computing environment 100 includes, for example, computer 101, wide area network (WAN) 102, end user device (EUD) 103, remote server 104, public cloud 105, and private cloud 106. In this embodiment, computer 101 includes processor set 110 (including processing circuitry 120 and cache 121), communication fabric 111, volatile memory 112, persistent storage 113 (including operating system 122 and block 200, as identified above), peripheral device set 114 (including user interface (UI) device set 123, storage 124, and Internet of Things (IoT) sensor set 125), and network module 115. Remote server 104 includes remote database 130. Public cloud 105 includes gateway 140, cloud orchestration module 141, host physical machine set 142, virtual machine set 143, and container set 144.

[0027] COMPUTER 101 may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database 130. As is well

understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment 100, detailed discussion is focused on a single computer, specifically computer 101, to keep the presentation as simple as possible. Computer 101 may be located in a cloud, even though it is not shown in a cloud in FIG. 1. On the other hand, computer 101 is not required to be in a cloud except to any extent as may be affirmatively indicated.

[0028] PROCESSOR SET 110 includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry 120 may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry 120 may implement multiple processor threads and/or multiple processor cores. Cache 121 is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set 110. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set 110 may be designed for working with qubits and performing quantum computing.

[0029] Computer readable program instructions are typically loaded onto computer 101 to cause a series of operational steps to be performed by processor set 110 of computer 101 and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache 121 and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set 110 to control and direct performance of the inventive methods. In computing environment 100, at least some of the instructions for performing the inventive methods may be stored in block 200 in persistent storage 113.

[0030] COMMUNICATION FABRIC 111 is the signal conduction path that allows the various components of computer 101 to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up buses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

[0031] VOLATILE MEMORY 112 is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, volatile memory 112 is characterized by random access, but this is not required unless affirmatively indicated. In computer 101, the volatile memory 112 is located in a single package and is internal to computer 101, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer 101.

[0032] PERSISTENT STORAGE **113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface-type operating systems that employ a kernel. The code included in block **200** typically includes at least some of the computer code involved in performing the inventive methods.

[0033] PERIPHERAL DEVICE SET **114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion-type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (for example, where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

[0034] NETWORK MODULE **115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from

an external computer or external storage device through a network adapter card or network interface included in network module **115**.

[0035] WAN **102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN **102** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

[0036] END USER DEVICE (EUD) **103** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

[0037] REMOTE SERVER **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

[0038] PUBLIC CLOUD **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages the transfer and storage of images, deploys new instantiations of VCEs

and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

[0039] Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

[0040] PRIVATE CLOUD **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

[0041] Measured service: cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, reported, and invoiced, providing transparency for both the provider and consumer of the utilized service.

[0042] With reference to FIG. 2, this figure depicts a block diagram of an example configuration for recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment. Application **200** is the same as application **200** in FIG. 1.

[0043] In the illustrated embodiment, tone extraction module **210** receives a plurality of voice samples of a user speaking in a specified voice tone. Module **210** uses a presently available technique to extract voice tone data from the plurality of voice samples. The voice tone data includes data usable to generate the specified voice tone. Some non-limiting examples of presently available techniques to extract voice tone data are a voice pitch analyzer which determines voice pitch (i.e., frequency) and variations in pitch, and machine learning models trained to identify emotions such as happiness, sadness, anger, or excitement within a plurality of voice samples. A user has the option to save or discard voice tone data usable to generate a specified voice tone, to associate a specific recipient for a specific

saved voice tone, and to set a default saved voice tone. For example, module **210** might receive a plurality of voice samples of a user speaking in a warm voice tone, extract voice tone data from the voice samples, allow the user to save the voice tone data as “family”, and associate the “family” voice tone data with contact information for the user’s spouse. As another example, module **210** might receive a plurality of voice samples of a user speaking in a cool, formal voice tone, extract voice tone data from the voice samples, allow the user to save the voice tone data as “customer”, and designate the “customer” voice tone data as the default voice tone. One implementation of module **210** saves, maintains, and manages voice tone data in a voice tone repository specific to a user.

[0044] Tone selection module **220** offers a user an option to select a saved voice tone for use during a voice communication (e.g., an incoming or outgoing audio call or audio or video conferencing session). A user saves a user’s voice tone selection for use in an ongoing voice communication, as well as in future voice communications with the same recipient as the current voice communication. For example, if the user selects the saved “family” voice tone for use in a call with a device designated in the user’s contacts as the user’s spouse, module **220** uses the saved “family” voice tone in the user’s next voice communication with his or her spouse. Another implementation of module **220** selects a saved voice tone for use during a particular voice communication automatically, using a previously-saved voice tone or a default voice tone.

[0045] During a voice communication (e.g., an audio call or audio or video conferencing session) in which a saved voice tone has been selected, tone adjustment module **230** uses a presently available speech to text model to convert the user’s speech input to corresponding text, and generates a speech output corresponding to the text. The speech output includes audio generated from the text using a presently available text to speech model and a voice tone generated using the saved voice tone data and a presently available technique. The intermediate text is necessary as currently available voice tone generation techniques operate on text rather than speech in audio form.

[0046] With reference to FIG. 3, this figure depicts an example of recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment. The example can be executed using application **200** in FIG. 2. Tone extraction module **210**, tone selection module **220**, and tone adjustment module **230** are the same as tone extraction module **210**, tone selection module **220**, and tone adjustment module **230** in FIG. 2.

[0047] As depicted, tone extraction module **210** receives voice sample(s) **300**, a plurality of voice samples of a user speaking in a specified voice tone. Module **210** uses a presently available technique to extract voice tone data **311** from the plurality of voice samples. Voice tone data **311** includes data usable to generate the specified voice tone. Module **210** stores voice tone data **311** in tone repository **310**.

[0048] Using a source of incoming call **320** (an example voice communication) and tone history **325** (including a user’s previous voice tone selection for the source of incoming call **320**), tone selection module **220** selects tone selection **330** for use with incoming call **320**. Tone selection **330** includes voice tone data **311**. Thus, tone adjustment module **230** converts user speech input **340** to corresponding text,

and generates speech output **350** corresponding to the text. Speech output **350** includes audio generated from the text using a presently available text to speech model and a voice tone generated using a voice tone data **311**.

[0049] With reference to FIG. 4, this figure depicts a flowchart of an example process for recipient-specific voice tone adjustment in telephony in accordance with an illustrative embodiment. Process **400** can be implemented in application **200** in FIG. 2.

[0050] In the illustrated embodiment, at block **402**, the process extracts, from a plurality of voice samples, voice tone data. At block **404**, the process uses a speech to text model, convert a speech input to corresponding text. At block **406**, the process generates a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data. Then the process ends.

[0051] The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

[0052] Additionally, the term “illustrative” is used herein to mean “serving as an example, instance or illustration.” Any embodiment or design described herein as “illustrative” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “at least one” and “one or more” are understood to include any integer number greater than or equal to one, i.e., one, two, three, four, etc. The terms “a plurality” are understood to include any integer number greater than or equal to two, i.e., two, three, four, five, etc. The term “connection” can include an indirect “connection” and a direct “connection.”

[0053] References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may or may not include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0054] The terms “about,” “substantially,” “approximately,” and variations thereof, are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of +8% or 5%, or 2% of a given value.

[0055] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the

described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

[0056] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

[0057] Thus, a computer implemented method, system or apparatus, and computer program product are provided in the illustrative embodiments for managing participation in online communities and other related features, functions, or operations. Where an embodiment or a portion thereof is described with respect to a type of device, the computer implemented method, system or apparatus, the computer program product, or a portion thereof, are adapted or configured for use with a suitable and comparable manifestation of that type of device.

[0058] Where an embodiment is described as implemented in an application, the delivery of the application in a Software as a Service (SaaS) model is contemplated within the scope of the illustrative embodiments. In a SaaS model, the capability of the application implementing an embodiment is provided to a user by executing the application in a cloud infrastructure. The user can access the application using a variety of client devices through a thin client interface such as a web browser (e.g., web-based e-mail), or other light-weight client-applications. The user does not manage or control the underlying cloud infrastructure including the network, servers, operating systems, or the storage of the cloud infrastructure. In some cases, the user may not even manage or control the capabilities of the SaaS application. In some other cases, the SaaS implementation of the application may permit a possible exception of limited user-specific application configuration settings.

[0059] Embodiments of the present invention may also be delivered as part of a service engagement with a client corporation, nonprofit organization, government entity, internal organizational structure, or the like. Aspects of these embodiments may include configuring a computer system to perform, and deploying software, hardware, and web services that implement, some or all of the methods described herein. Aspects of these embodiments may also include analyzing the client's operations, creating recommendations responsive to the analysis, building systems that implement portions of the recommendations, integrating the systems into existing processes and infrastructure, metering use of the systems, allocating expenses to users of the systems, and billing for use of the systems. Although the above embodiments of present invention each have been described by stating their individual advantages, respectively, present invention is not limited to a particular combination thereof. To the contrary, such embodiments may also be combined in

any way and number according to the intended deployment of present invention without losing their beneficial effects.

What is claimed is:

1. A computer-implemented method comprising:
 - extracting, from a plurality of voice samples, voice tone data;
 - converting, using a speech to text model, a speech input to corresponding text; and
 - generating a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data.
2. The computer-implemented method of claim 1, wherein the voice tone data comprises data usable to generate the voice tone.
3. The computer-implemented method of claim 1, wherein the voice tone data is maintained in a user-specific voice tone repository.
4. The computer-implemented method of claim 1, further comprising:
 - selecting, for use in a voice communication with a communication recipient, the voice tone data.
5. The computer-implemented method of claim 4, wherein the voice tone data was previously selected for use in a previous voice communication with the communication recipient.
6. The computer-implemented method of claim 4, wherein the voice tone data is default voice tone data.
7. A computer program product comprising one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media, the program instructions executable by a processor to cause the processor to perform operations comprising:
 - extracting, from a plurality of voice samples, voice tone data;
 - converting, using a speech to text model, a speech input to corresponding text; and
 - generating a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data.
8. The computer program product of claim 7, wherein the stored program instructions are stored in a computer readable storage device in a data processing system, and wherein the stored program instructions are transferred over a network from a remote data processing system.
9. The computer program product of claim 7, wherein the stored program instructions are stored in a computer readable storage device in a server data processing system, and wherein the stored program instructions are downloaded in response to a request over a network to a remote data

processing system for use in a computer readable storage device associated with the remote data processing system, further comprising:

- program instructions to meter use of the program instructions associated with the request; and
 - program instructions to generate an invoice based on the metered use.
10. The computer program product of claim 7, wherein the voice tone data comprises data usable to generate the voice tone.
 11. The computer program product of claim 7, wherein the voice tone data is maintained in a user-specific voice tone repository.
 12. The computer program product of claim 7, further comprising:
 - selecting, for use in a voice communication with a communication recipient, the voice tone data.
 13. The computer program product of claim 12, wherein the voice tone data was previously selected for use in a previous voice communication with the communication recipient.
 14. The computer program product of claim 12, wherein the voice tone data is default voice tone data.
 15. A computer system comprising a processor and one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media, the program instructions executable by the processor to cause the processor to perform operations comprising:
 - extracting, from a plurality of voice samples, voice tone data;
 - converting, using a speech to text model, a speech input to corresponding text; and
 - generating a speech output corresponding to the text, the speech output comprising audio generated from the text using a text to speech model and a voice tone generated using the voice tone data.
 16. The computer system of claim 15, wherein the voice tone data comprises data usable to generate the voice tone.
 17. The computer system of claim 15, wherein the voice tone data is maintained in a user-specific voice tone repository.
 18. The computer system of claim 15, further comprising:
 - selecting, for use in a voice communication with a communication recipient, the voice tone data.
 19. The computer system of claim 18, wherein the voice tone data was previously selected for use in a previous voice communication with the communication recipient.
 20. The computer system of claim 18, wherein the voice tone data is default voice tone data.

* * * * *